

Read

Process

Write

```
streaming_df = spark.readStream
```

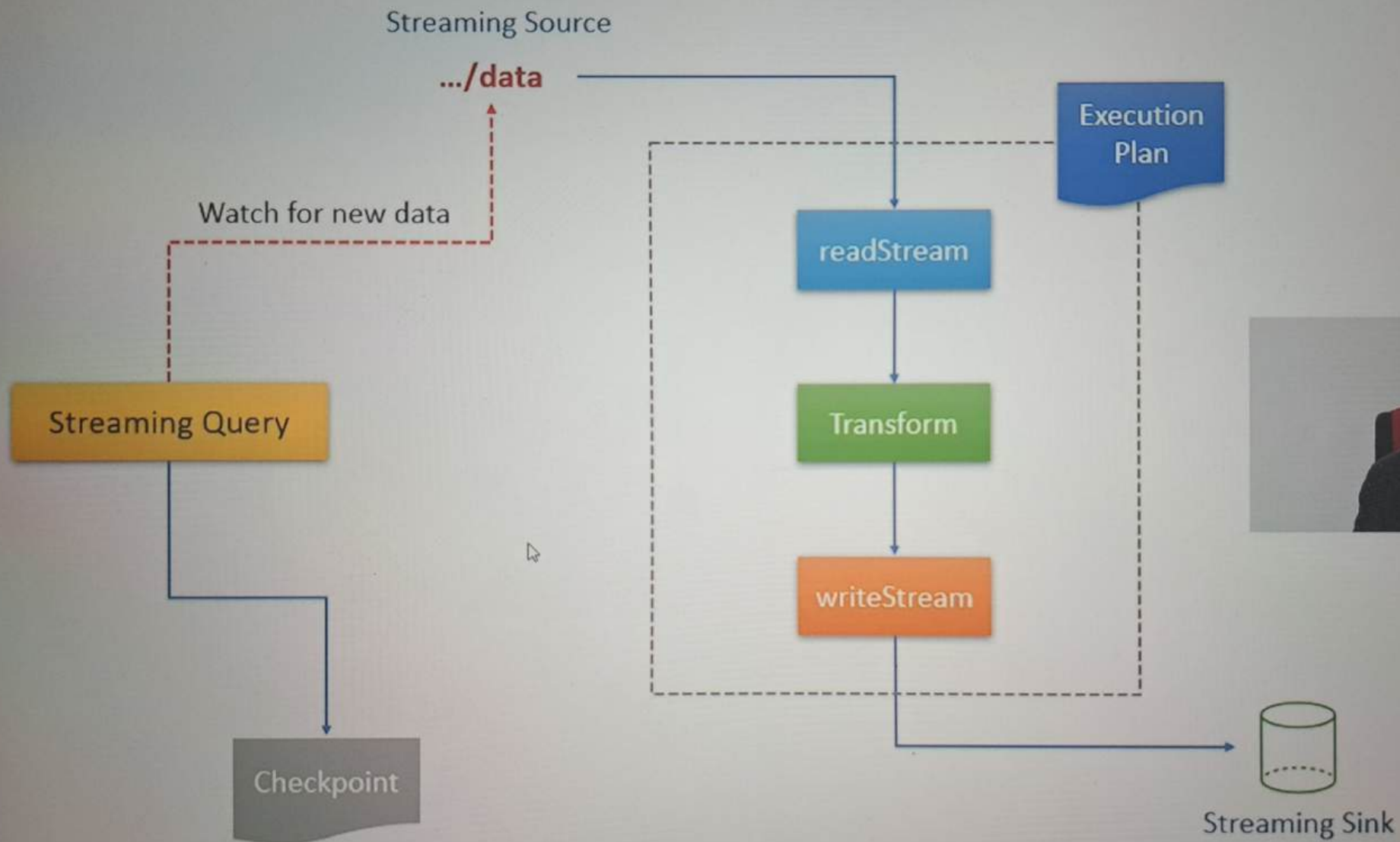
```
result_df = streaming_df.select(...)  
                        .where(...)  
                        .groupBy(...)  
                        .agg(...)
```

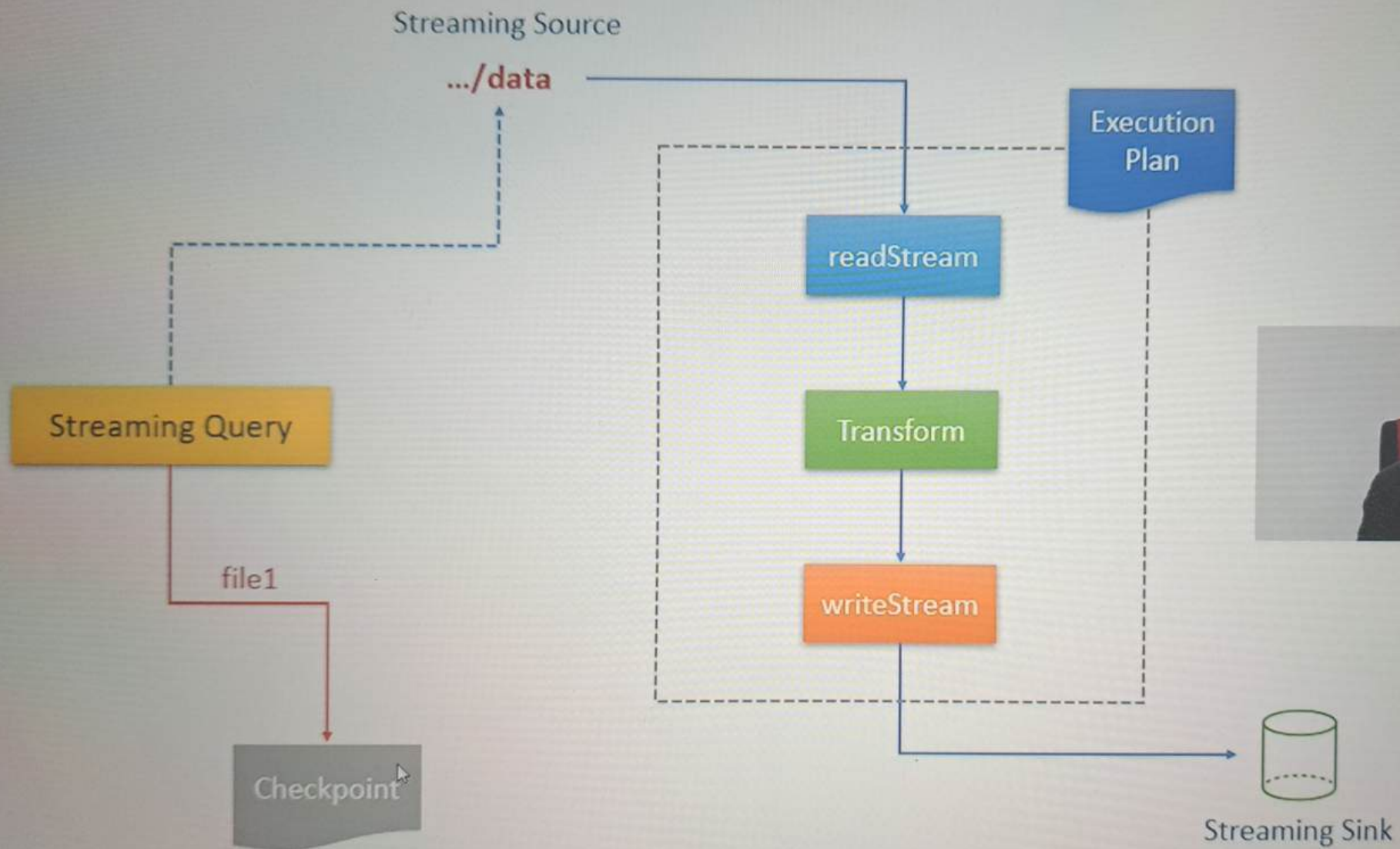
```
result_df.writeStream  
    .option("checkpointLocation", ....)  
    .outputMode(...)  
    .toTable(...)
```

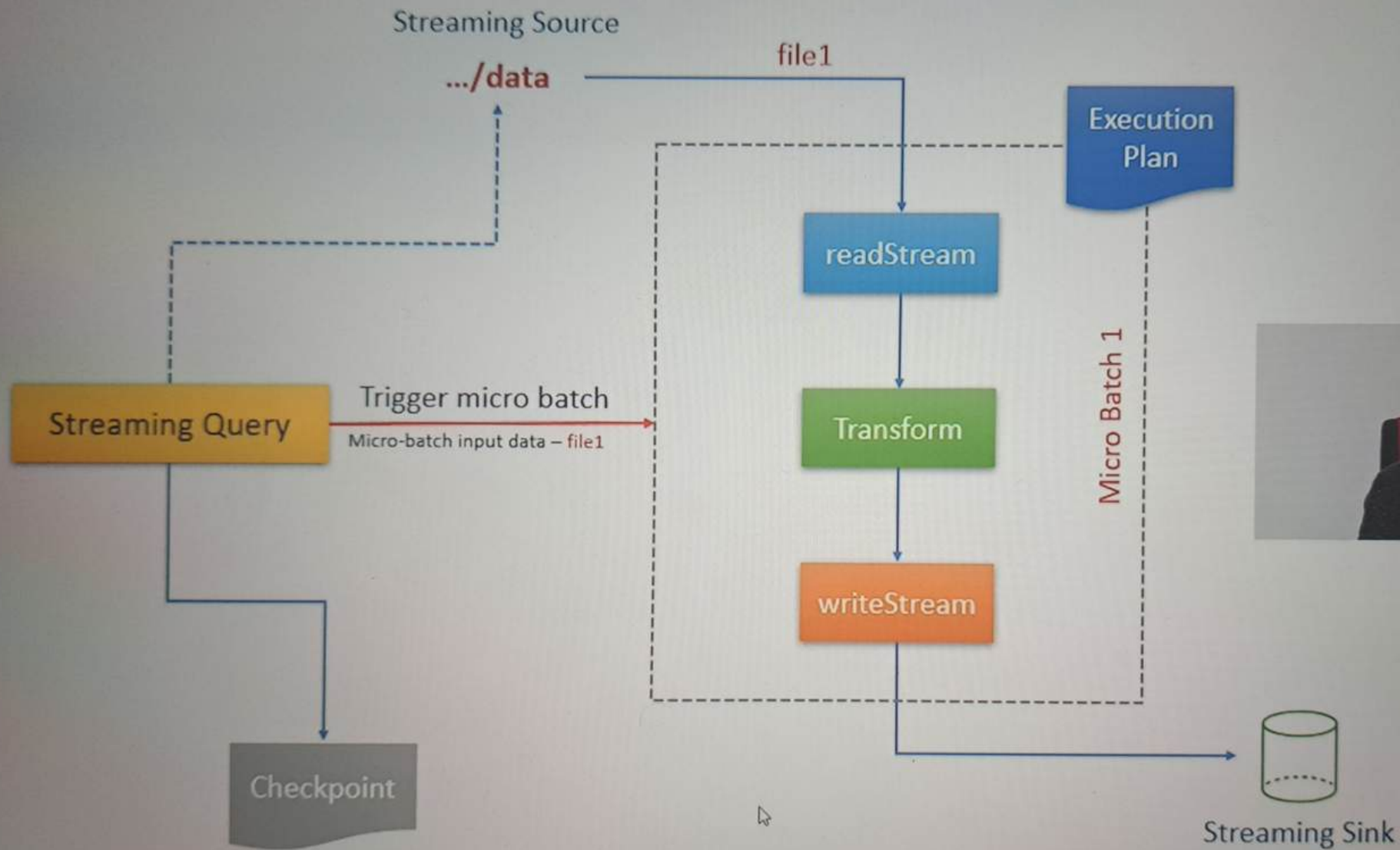
Spark
SQL Engine

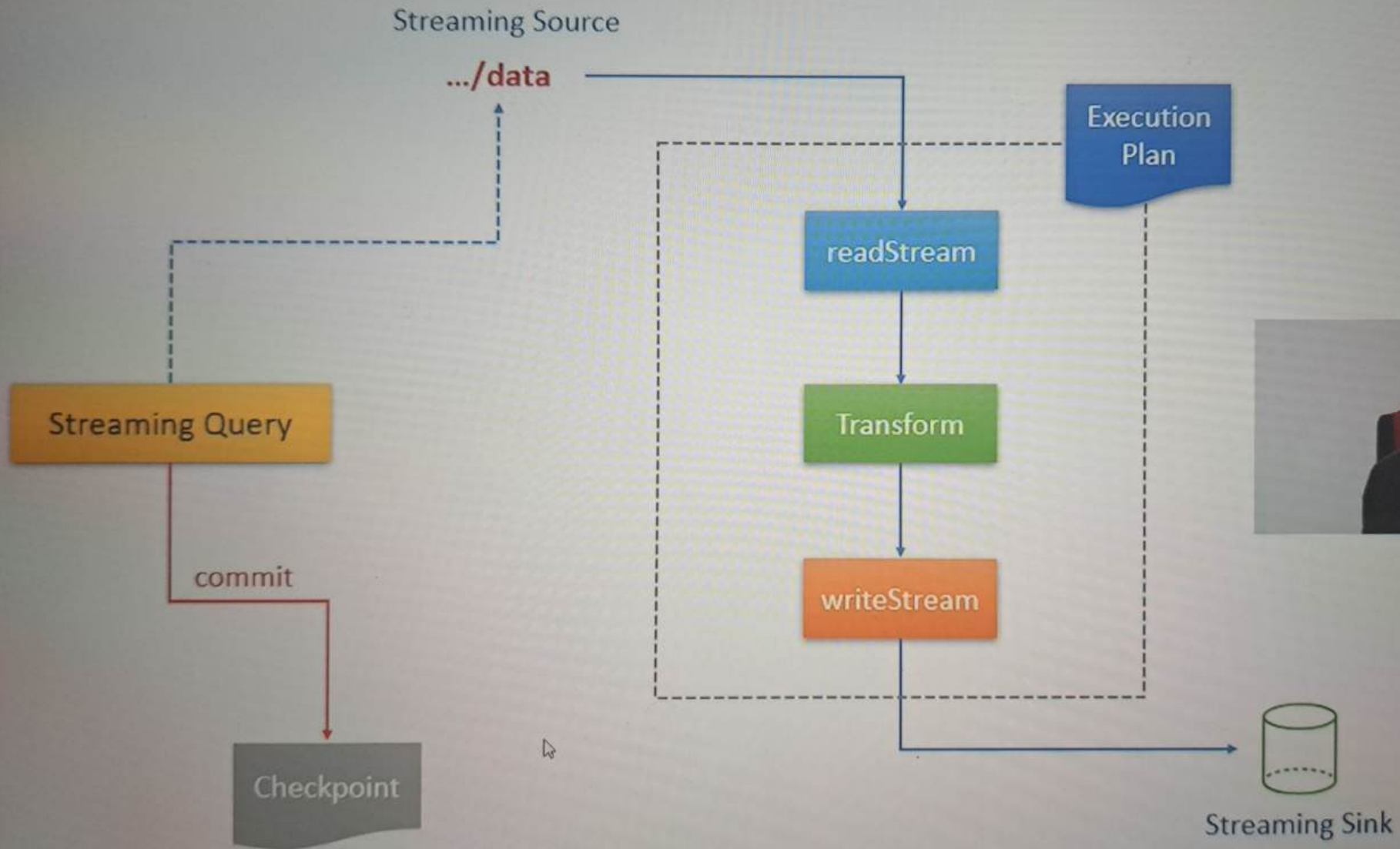
Execution
Plan

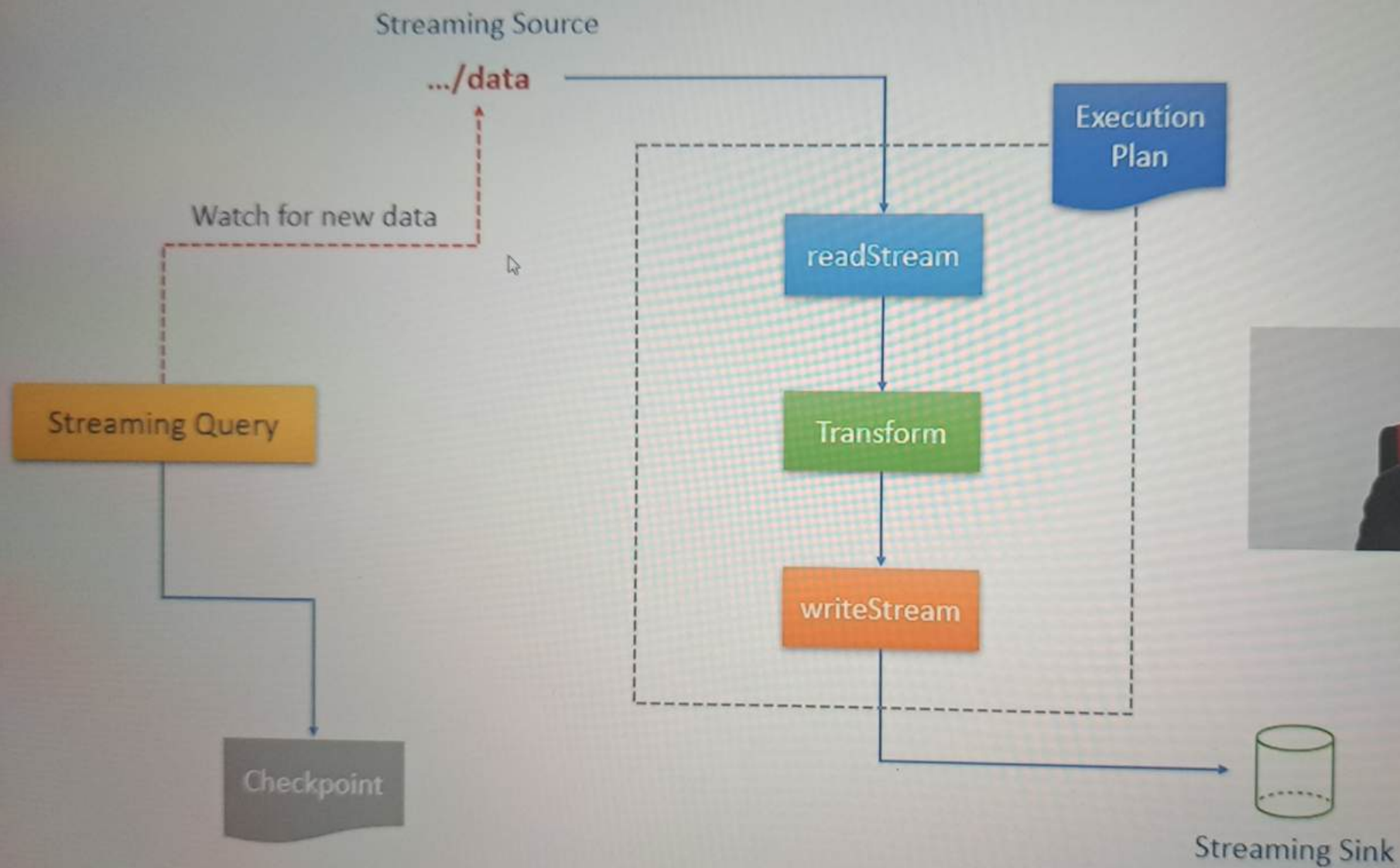


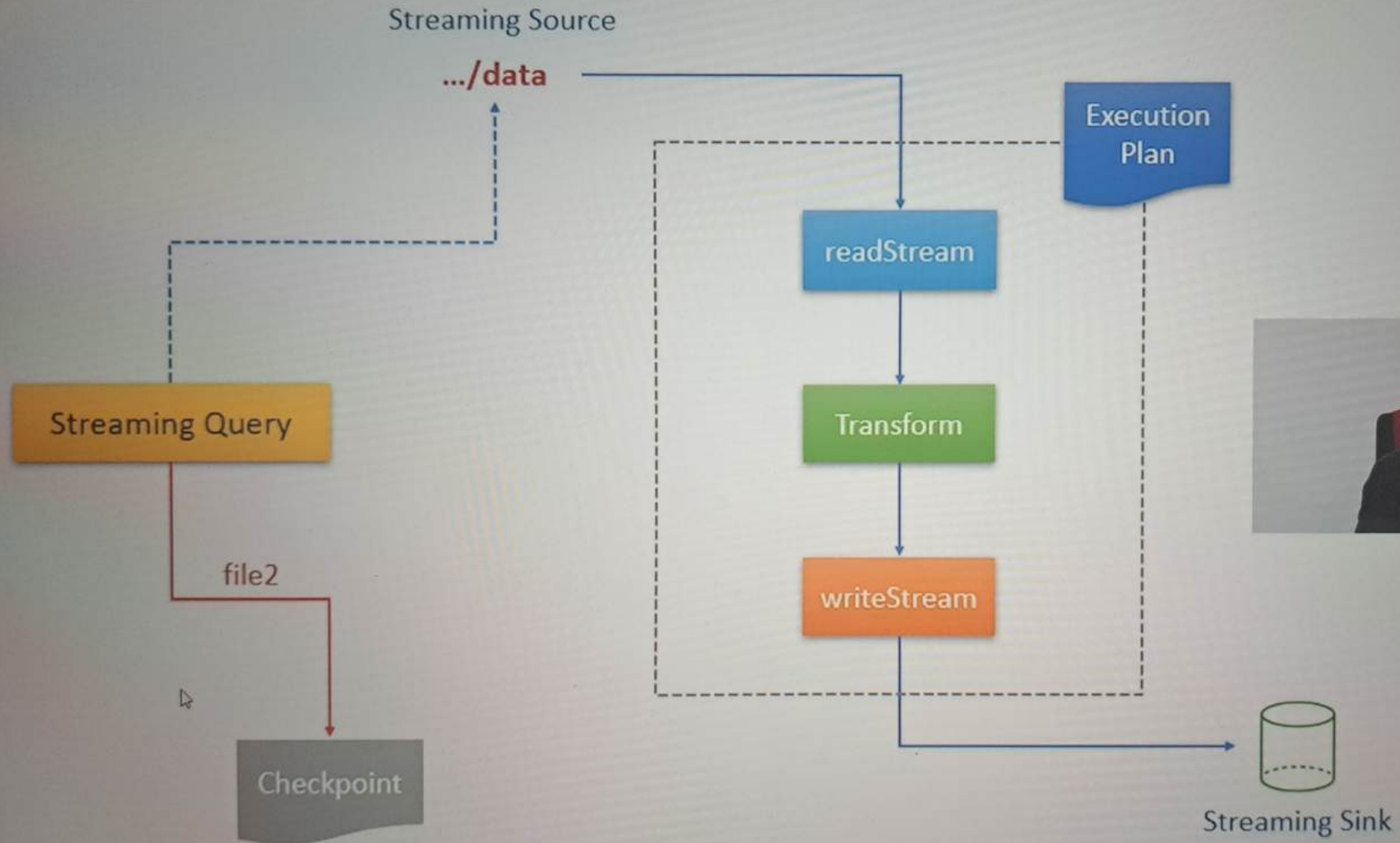


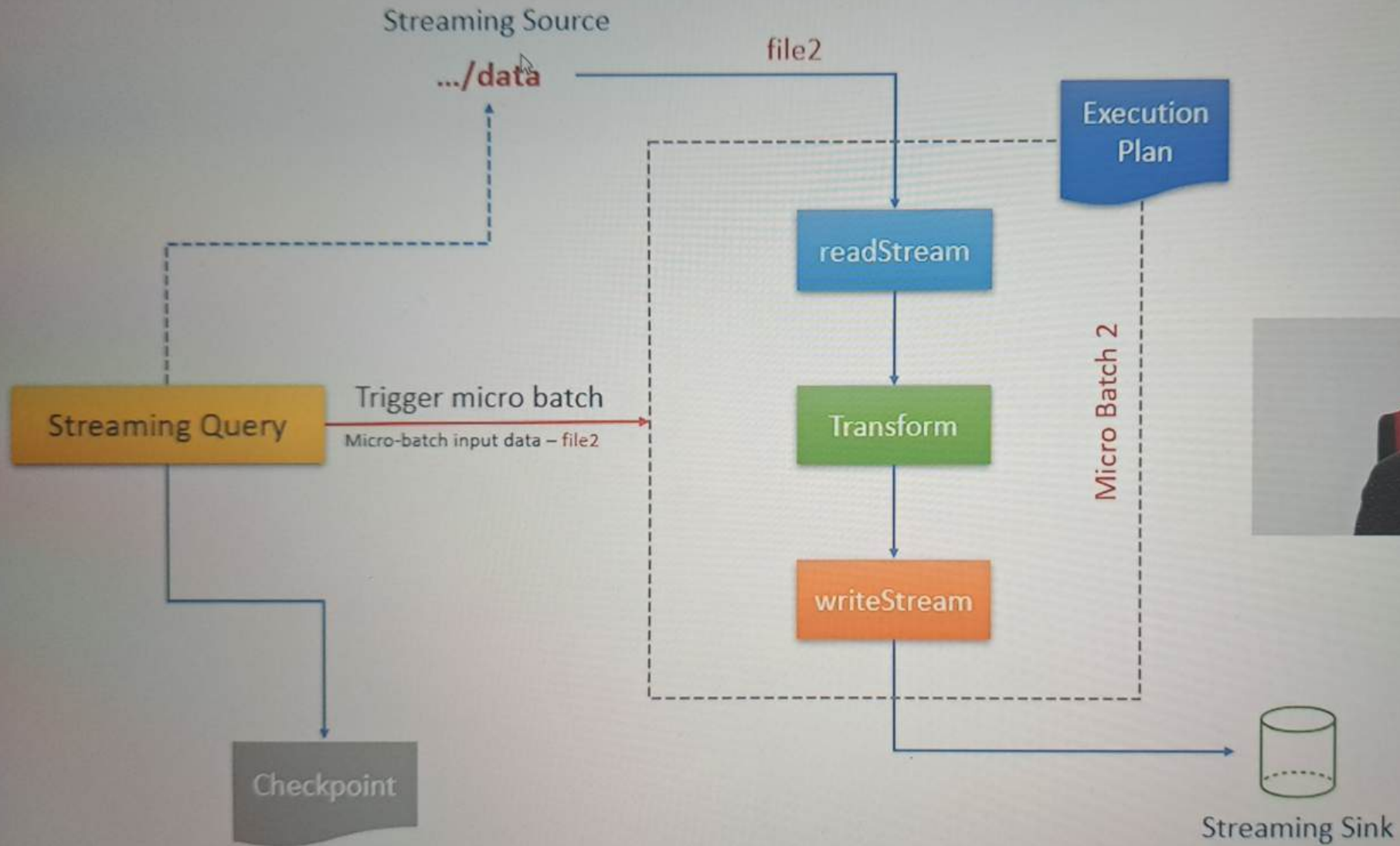


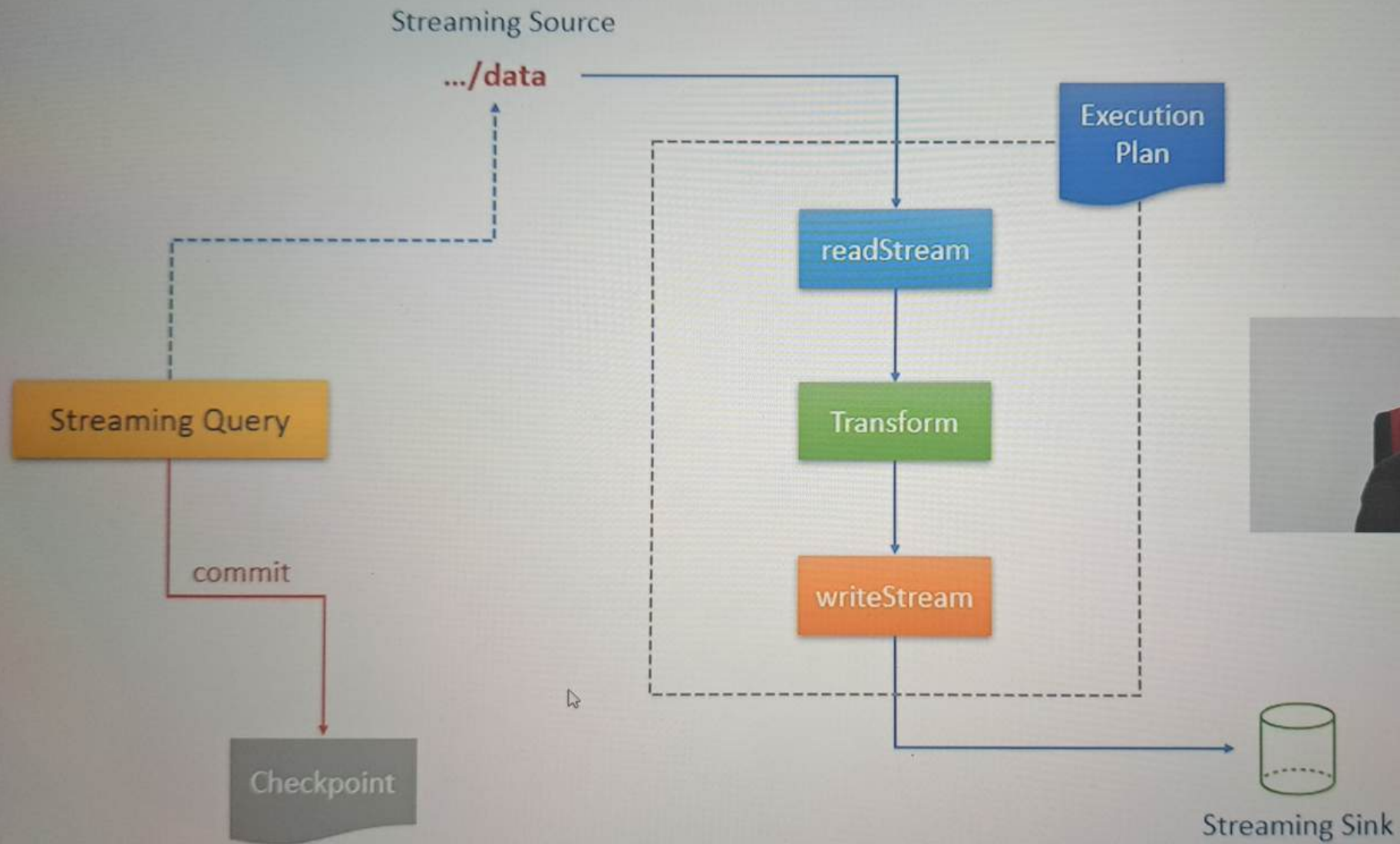


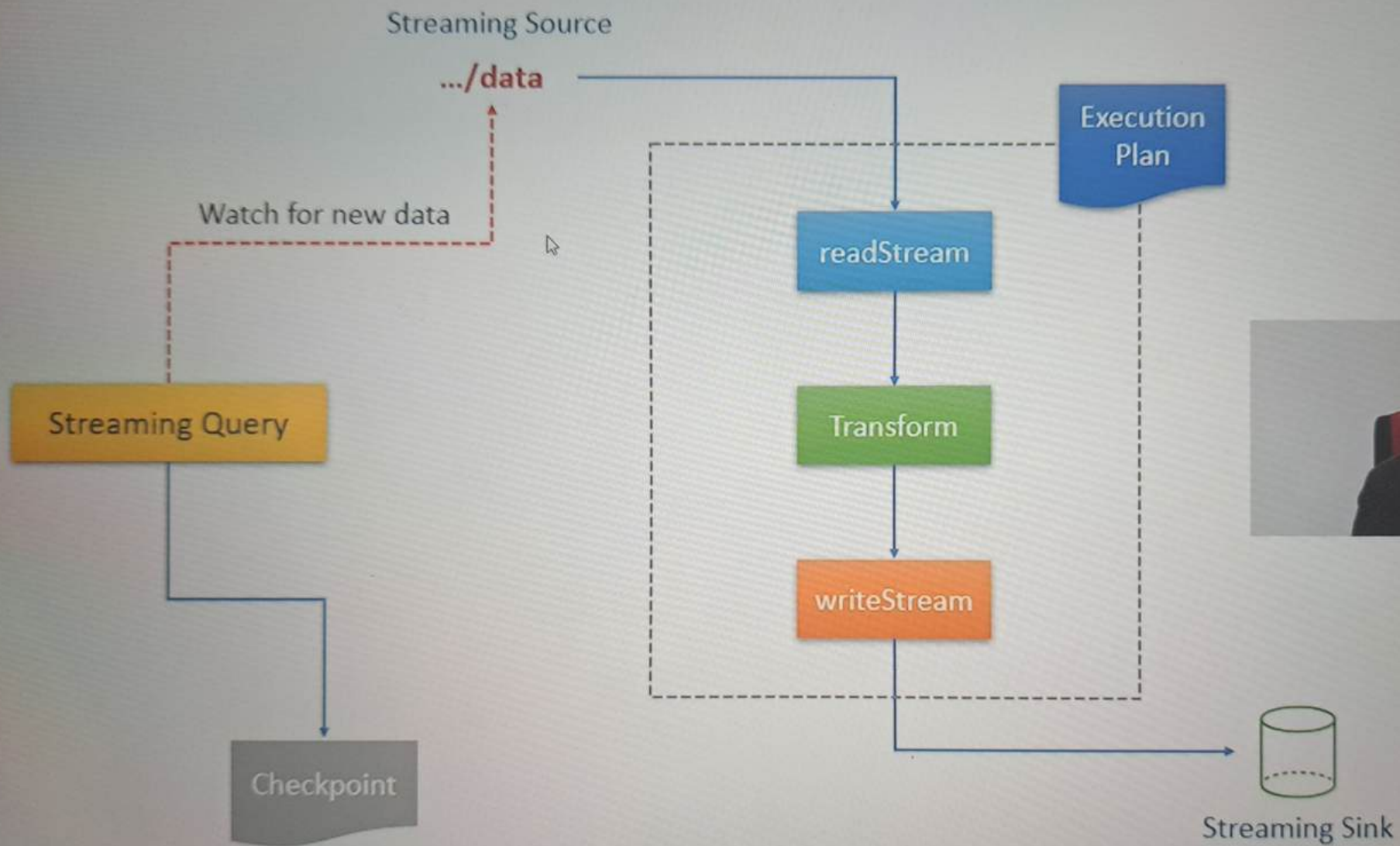


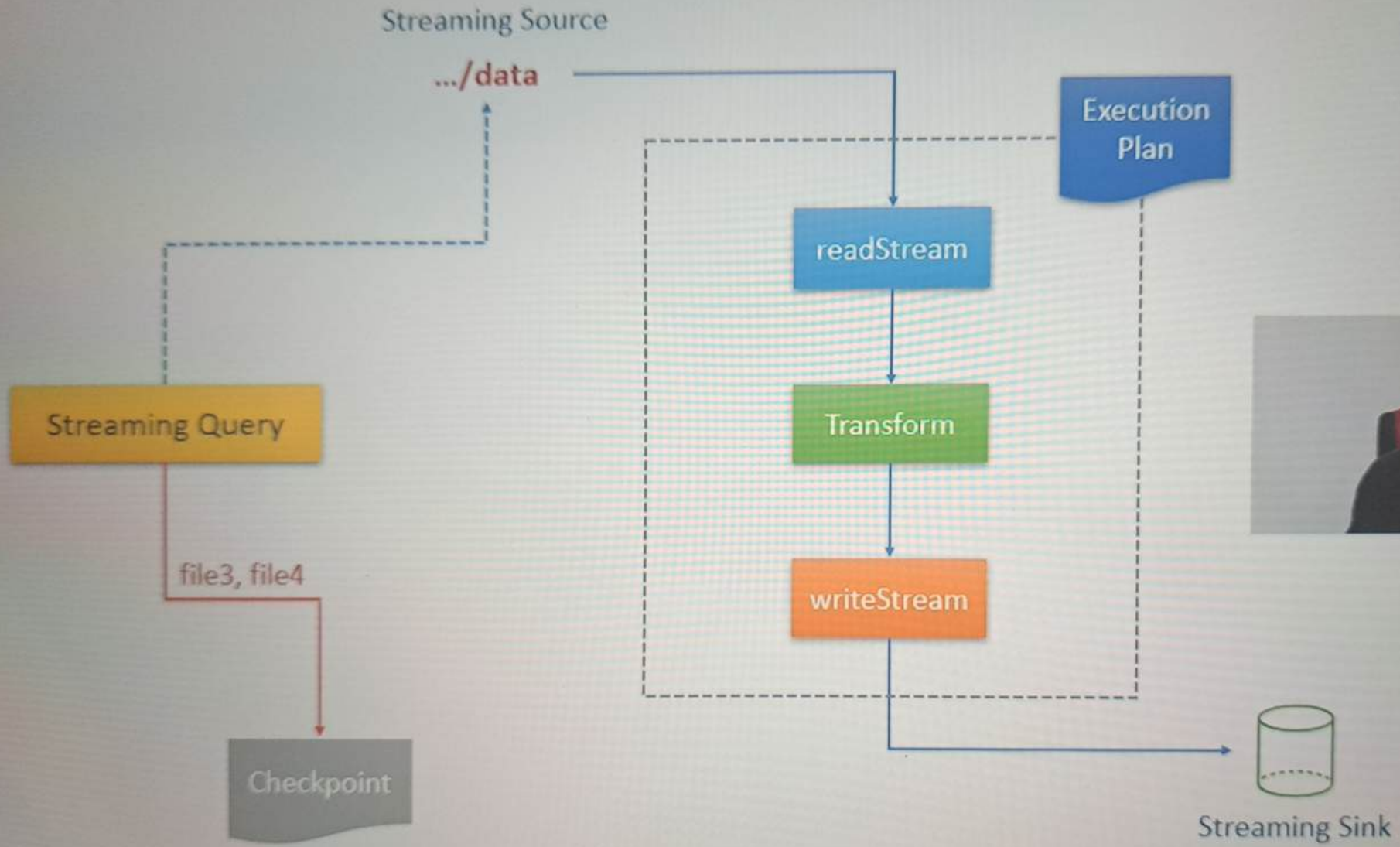


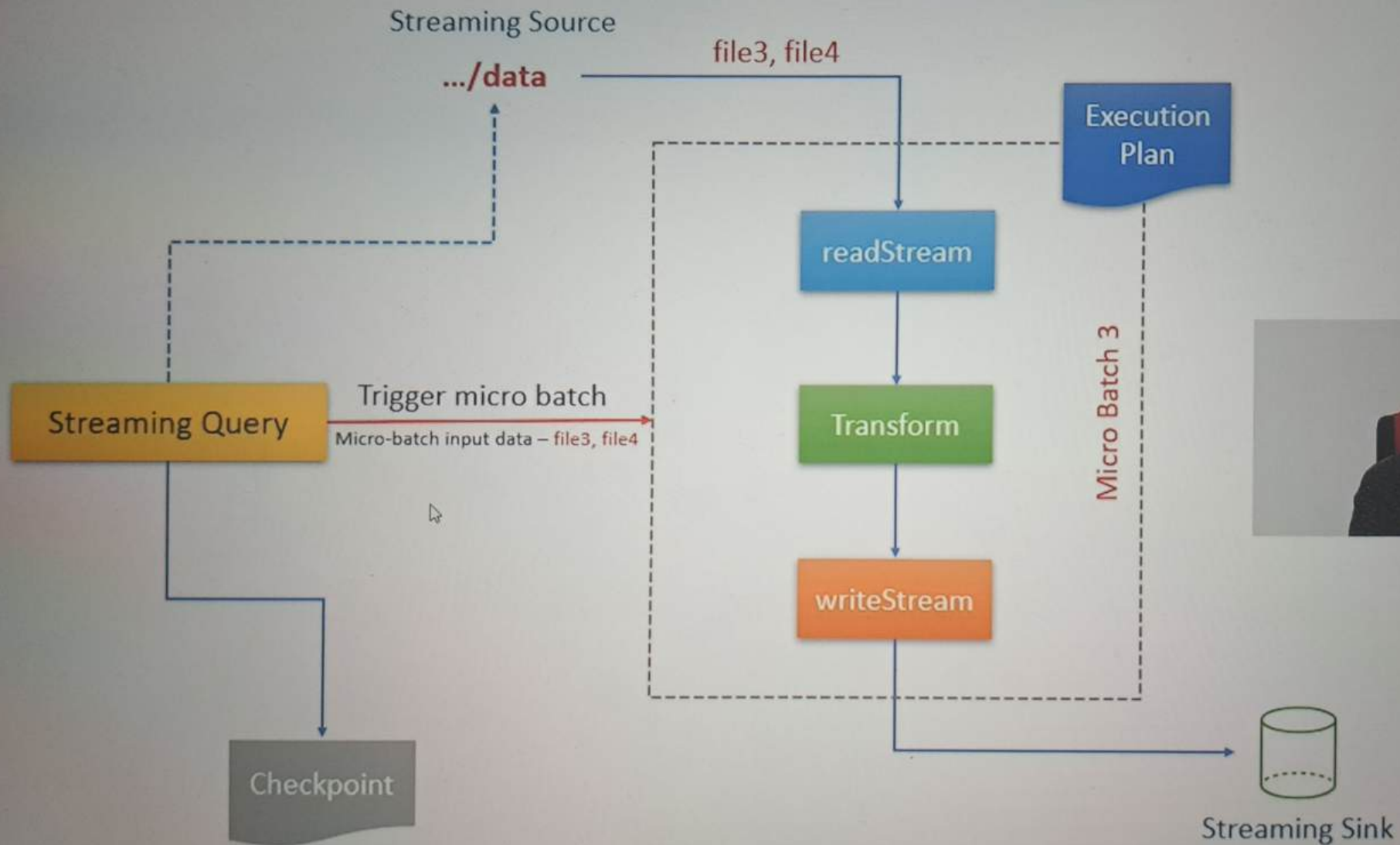












APACHE
SparkTM
Structured Streaming



Streaming Query – A background thread (coordinator) to manage the execution of a streaming job



Micro batch – Streaming query runs micro-batches in a loop, and each batch is initialized with a batch dataset



Checkpoint – Checkpoint contains the information about micro-batch data



Implicitly Incremental – Streaming Query ensures reading only the new data in each micro batch